Static Code Analysis Report

MESample v1.3.0 — clang-tidy

Software	MESample 1.3.0
Tool	clang-tidy (LLVM)
Tool Version	21.0.0 (Android NDK 29.0.14206865 toolchain)
Analysis Mode	Compilation-database based (aarch64-linux-android24, C++20)
Check Sets	clang-analyzer-*, bugprone-*, performance-*, portability-*, readability-*, modernize-*, cppcoreguidelines-*
Scope	Self-developed C++ source (9 files: register_types.cpp, VehicleData.cpp + headers)
Date	2026-09-09

1. Results Overview

Category	Count	Severity
Security defects (clang-analyzer)	**0**	**Pass**
Potential bug patterns (bugprone)	17	**Low**
Performance suggestions	19	Info
Best-practice / style (modernize, readability, cppcoreguidelines)	416	Info
Total diagnostics	**452**	—

2. Notable Findings (bugprone, non-security)

File	Issue	Category
DoorController.h	Assignment within an `if` condition	bugprone-assignment-in-if-condition
LampController.h	Macro arguments not enclosed in parentheses (3)	bugprone-macro-parentheses

CurveMover3D.h	`operator=` does not handle self-assignment	bugprone-unhandled-self-assignment
CurveMover3D.h, VehicleData.cpp	Narrowing conversions (double→float, int64→int)	bugprone-narrowing-conversions
Funclib.h	Unused local non-trivial variables (6)	bugprone-unused-local-non-trivial-variable

3. Conclusion

Item	Status
Security vulnerabilities	**PASS — 0 security defects**

All findings are code-quality issues common in demo/sample code; none constitute a security vulnerability. The project is compiled with `-Wall -Wextra -Werror`, guaranteeing zero compiler warnings.

Report generated: 2026-09-09 | Tool: clang-tidy 21.0.0 | MESample v1.3.0